WORKED SOLUTION

STEP I 2003, Question 3

Pure • Half-Angle Trigonometric Equations

STEP I 2003, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q3

The key idea

Each part compares a function of $\frac{1}{2}\theta$ with the same function of θ, so write everything in terms of the half-angle using a double-angle formula. Each equation then becomes a polynomial equation in a single quantity ($\sin\frac{1}{2}\theta$, $\tan\frac{1}{2}\theta$ or $\cos\frac{1}{2}\theta$), which we can factorise or solve as a quadratic. The only care needed is to keep track of which values the half-angle function can actually take, and where the functions are defined.

Solution

Part (i)

Using $\sin\theta = 2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$, the equation $2\sin\frac{1}{2}\theta = \sin\theta$ is equivalent to

$$2\sin\tfrac{1}{2}\theta - 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta = 0, \qquad \text{that is,} \qquad 2\sin\tfrac{1}{2}\theta\,(1-\cos\tfrac{1}{2}\theta) = 0.$$

So the equation holds if and only if $\sin\frac{1}{2}\theta = 0$ or $\cos\frac{1}{2}\theta = 1$.

But if $\cos\frac{1}{2}\theta = 1$ then $\sin^2\frac{1}{2}\theta = 1 - \cos^2\frac{1}{2}\theta = 0$, so $\sin\frac{1}{2}\theta = 0$ anyway. The second possibility is therefore contained in the first, and

$$2\sin\tfrac{1}{2}\theta = \sin\theta \iff \sin\tfrac{1}{2}\theta = 0,$$

as required. (In other words, the solutions are exactly $\theta = 2n\pi$ for integer n.)

Part (ii)

Let $t = \tan\frac{1}{2}\theta$. For the equation to make sense we need both $\tan\frac{1}{2}\theta$ and $\tan\theta$ to be defined, so $\cos\frac{1}{2}\theta \neq 0$ and $\cos\theta \neq 0$. The second condition means $t^2 \neq 1$, since $\cos\theta = \dfrac{1-t^2}{1+t^2}$. With these restrictions, $\tan\theta = \dfrac{2t}{1-t^2}$, and the equation becomes

$$2t = \frac{2t}{1-t^2}.$$

Multiplying by $1 - t^2$, which is non-zero, gives $2t - 2t^3 = 2t$, so $t^3 = 0$ and $t = 0$.

Ask yourself Is it safe to multiply through by $1 - t^2$, and have we lost or gained any solutions?

Multiplying by a non-zero quantity is reversible, so the equation in t is exactly equivalent to $t = 0$ on the set where both tangents are defined. Conversely, $t = 0$ gives $\tan\theta = 0$ and $\tan\frac{1}{2}\theta = 0$, and both sides are 0.

Now $\tan\frac{1}{2}\theta = 0$ exactly when $\frac{1}{2}\theta = n\pi$. So the solutions are

$$\theta = 2n\pi, \qquad n \in \mathbb{Z},$$

the same set as in part (i).

Part (iii)

Let $c = \cos\frac{1}{2}\theta$. Using $\cos\theta = 2\cos^2\frac{1}{2}\theta - 1 = 2c^2 - 1$, the equation $2\cos\frac{1}{2}\theta = \cos\theta$ is equivalent to

$$2c = 2c^2 - 1 \iff 2c^2 - 2c - 1 = 0 \iff c = \frac{2 \pm \sqrt{4+8}}{4} = \frac{1 \pm \sqrt{3}}{2}.$$

Since c is a cosine, $-1 \le c \le 1$. The root $\frac{1}{2}(1+\sqrt{3}) \approx 1.37$ is too big, while $\frac{1}{2}(1-\sqrt{3}) \approx -0.37$ is allowed. So the equation holds if and only if

$$\cos\tfrac{1}{2}\theta = \tfrac{1}{2}(1-\sqrt{3}) = -\cos\phi,$$

where ϕ is as defined in the question. This ϕ exists and is unique because $0 < \frac{1}{2}(\sqrt{3}-1) < 1$ and cos decreases from 1 to 0 on $[0, \frac{1}{2}\pi]$.

Ask yourself How do we turn $\cos\frac{1}{2}\theta = -\cos\phi$ into the given form?

Write $-\cos\phi = \cos(\pi - \phi)$. The general solution of $\cos x = \cos\alpha$ is $x = 2m\pi \pm \alpha$, so

$$\tfrac{1}{2}\theta = 2m\pi \pm (\pi - \phi) \qquad \text{that is,} \qquad \theta = 4m\pi \pm (2\pi - 2\phi), \quad m \in \mathbb{Z}.$$

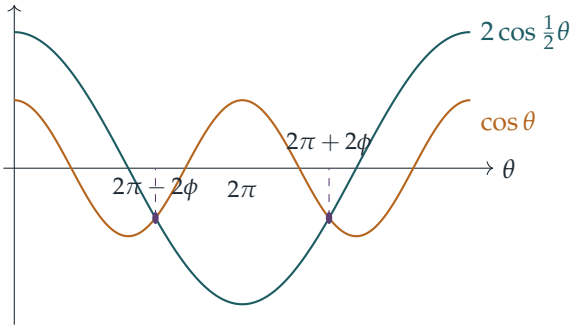
Take the two signs separately.

- With the $+$ sign: $\theta = 4m\pi + 2\pi - 2\phi = (4n+2)\pi - 2\phi$ with $n = m$.
- With the $-$ sign: $\theta = 4m\pi - 2\pi + 2\phi = \big(4(m-1)+2\big)\pi + 2\phi = (4n+2)\pi + 2\phi$ with $n = m - 1$.

As m runs over all integers, so does n in each case. Hence

$$2\cos\tfrac{1}{2}\theta = \cos\theta \iff \theta = (4n+2)\pi \pm 2\phi \text{ for some integer } n.$$

Every step was an equivalence, so both directions are proved. As a direct check of the converse: if $\theta = (4n+2)\pi \pm 2\phi$ then $\cos\frac{1}{2}\theta = \cos\big((2n+1)\pi \pm \phi\big) = -\cos\phi = c$, which satisfies $2c = 2c^2 - 1$.



Over one period $0 \leq \theta \leq 4\pi$ of $2\cos\frac{1}{2}\theta$, the curves meet twice, symmetrically about $\theta = 2\pi$, where $2\cos\frac{1}{2}\theta$ reaches its minimum of -2.

Checks

- Numerically $\cos\phi \approx 0.3660$ gives $\phi \approx 1.1961$, so the first positive solution of (iii) is $2\pi - 2\phi \approx 3.8911$. There $2\cos\frac{1}{2}\theta \approx -0.7321$ and $\cos\theta \approx -0.7321$, as required. A root search of $2\cos\frac{1}{2}\theta - \cos\theta$ on $[-6\pi, 6\pi]$ found exactly the six values $(4n+2)\pi \pm 2\phi$ in that range.
- A similar search for (ii), avoiding the poles, found roots only at multiples of 2π.
- Sanity check for (i) and (ii): $\theta = \pi$ fails (i), since $2\sin\frac{1}{2}\pi = 2 \neq 0 = \sin\pi$, and is outside the domain in (ii), where $\tan\frac{1}{2}\pi$ is undefined.

Common slips

- In (i), cancelling $\sin\frac{1}{2}\theta$ without saying it might be zero, or giving $\cos\frac{1}{2}\theta = 1$ as a separate family of solutions without noticing it is already included.
- In (ii), forgetting that θ must avoid the points where $\tan\theta$ or $\tan\frac{1}{2}\theta$ is undefined before clearing denominators.
- In (iii), keeping the root $\frac{1}{2}(1+\sqrt{3})$, which is bigger than 1, or mishandling the $\pm$ when converting $4m\pi \pm (2\pi - 2\phi)$ to the form $(4n+2)\pi \pm 2\phi$.

Beyond STEP

Each part solves an equation by substituting a single function of the half-angle, which turns a trigonometric equation into an algebraic one. The substitution $t = \tan\frac{1}{2}\theta$ in (ii) is especially powerful: it expresses $\sin\theta$, $\cos\theta$ and $\tan\theta$ as rational functions of t, which is why it is used in first-year calculus to integrate any rational function of $\sin\theta$ and $\cos\theta$. Geometrically it parametrises the unit circle by rational points. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with these ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.